Development of Autonomous Navigation System for Mobile Robots in Indoor Environment Using Navigation Beacon

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Nomenclature

- **AGV**: Automated Guided Vehicles
- **BLE**: Bluetooth Low Energy
- **DR**: Dead Reckoning (via wheel odometry)
- **DWA**: Dynamic Window Approach for local obstacle avoidance
- **EKF**: Extended Kalman Filter used for sensor fusion
- **FIM**: Fisher Information Matrix used in optimization frameworks
- **GNSS**: Global Navigation Satellite Systems
- **GPS**: Global Positioning System
- **HFOV**: Horizontal Field of View
- **IMU**: Inertial Measurement Unit
- **IPS**: Indoor Positioning System
- **LiDAR**: Light Detection and Ranging
- **RFID**: Radio Frequency Identification
- **ROS**: Robot Operating System
- **RRT**: Rapidly-exploring Random Trees
- **SLAM**: Simultaneous Localization and Mapping
- **TEB**: Time-Elastic Band algorithm for trajectory optimization
- **UWB**: Ultra-Wideband technology for localization

List of symbols:

- $\mathbf{A_j}$: Set of candidate tags used in mandatory placement constraints.
- $\mathbf{C}$ / $\mathbf{C_j}$: Candidate pool or specific candidate tags generated along wall surfaces.
- $\mathbf{d}$: Distance.
- $\mathbf{d_{calc}}$: Calculated distance used for feasibility analysis.
- $\mathbf{d_{max}}$: Maximum detection distance based on camera intrinsics and pixel thresholds.
- $\mathbf{f}$: Focal length of the camera.
- $\mathbf{f_x}$, $\mathbf{f_y}$: Camera intrinsics.
- $\mathbf{F}$: Functional tags placed at doors or task areas.
- $\mathbf{J}$: Junction tags placed at intersections.
- $\mathbf{M}$: Environment Map.
- $\mathbf{N_c}$: Total number of candidate tags.
- $\mathbf{N_p}$: Number of robot poses sampled along a path.
- $\mathbf{P}$ / $\mathbf{P_i}$: The robot's path or specific robot poses.
- $\mathbf{P_{min}}$ / **PxThr**: Pixel Thresholds for marker detection.
- $\mathbf{s}$: Tag size or dimensions.
- $\mathbf{S}$: The minimal subset of selected ArUco tags.
- $\mathbf{T_{final}}$: The final optimized set of tags.
- $\mathbf{V}$ / $\mathbf{V_{i,j}}$: Binary visibility matrix representing if a tag is visible from a specific pose.
- $\mathbf{x_j}$: Binary decision variable (1 if tag j is selected, 0 otherwise).
- Φ **(n)**: A saturation function ensuring coverage is valued over excessive redundancy.

ABSTRACT

Autonomous indoor navigation poses significant challenges, particularly in visually repetitive environments, such as hotel corridors, where Global Positioning System (GPS) signals are unavailable and wireless signals (e.g., Wi-Fi, BLE) are often unreliable due to multipath interference. While LiDAR-based systems offer high precision, their prohibitive cost and high computational complexity often limit accessibility for low-cost service robotics. The work presents a cost-effective vision-based navigation system utilizing ArUco markers as unique navigation beacons to resolve spatial ambiguity. ArUco tags were used as navigation beacons, classified, and placed based on their locations with the aim of maximizing localization availability while minimizing infrastructure redundancy. An indoor environment with repetitive door notches and junctions with varying corridor widths and lengths is created. A three-stage optimization algorithm that automates the transition from manual observation to a deterministic deployment strategy is introduced. First, a mathematical feasibility analysis selects optimal marker parameters based on camera intrinsics and environmental constraints. Second, a hierarchical placement logic categorizes markers into Functional, Junction, and Corridor tags, incorporating a greedy selection technique to determine the optimal placement and density of markers. The system is validated through MATLAB simulation, Webots simulation, and Hardware validation which achieved a localization availability of 94.2%, while reducing the total infrastructure by 82%, and a 100% pre-turn success rate at junctions and a mean visibility of 1.011 tags per frame thereby offering autonomous indoor navigation more accessible for real-world applications.

Keywords: Autonomous Indoor Navigation, ArUco Markers, Greedy Optimization, Vision-Based Localization, Mobile Robotics, Navigation Beacons.

CHAPTER I

INTRODUCTION

Indoor autonomous navigation is widely used in modern service robotics, particularly in structured environments like hotels, hospitals, and warehouses, and Indoor Positioning Systems (IPS) are essential for mobile robot navigation in these environments, where GPS are not available (Deshmukh et al., 2025). Outdoor robots rely on Global Navigation Satellite Systems (GNSS), Camera, Light Detection and Ranging (LiDAR), radar, ultrasonic, inertial measurement unit (IMU) and odometry sensors for perception applications. The most frequently used robot vision-based sensor fusion methods combine camera images with LiDAR point clouds. Research into sensor fusion techniques remains an important component of achieving better sensing capabilities for unstructured outdoor environments (Wijayathunga, Rassau and Chai, 2023).

The positioning problem in the outdoor environment has been solved by the ubiquitous global navigation satellite system (GNSS). However, due to the complex and diverse environmental conditions (factories, laboratories, hospitals, shopping malls, etc.), signal attenuation, multipath, and non-visual distance problems, the GNSS is no longer applicable in the indoor environment (Kim Geok et al., 2020). Therefore indoor positioning systems have been proposed by many researchers, where the mobile robot is allowed to complete tasks like pick and place, patrol, and take elevators (Abdulla et al., 2016). The high accuracy of indoor positioning is the premise to ensure that the mobile robot can complete the task and ensure the safety of the human–machine environment. The demand to achieve a centimetre-level positioning accuracy of mobile robots attracts researchers to continue to explore the positioning of indoor mobile robots (Huang et al., 2023).

For 2D localization, Laser range finders are far more precise and easier to work with than cameras (Vincent, Limketkai and Eriksen, 2010). Simultaneous localization and map building (SLAM) is not an effective method of localization for repetitive environments. LiDAR-based approaches, though highly accurate, significantly increase the system's computational load and hardware expenses, often making them inaccessible for low-cost service applications. Consequently, researchers have turned to alternative technologies such as Ultra-Wideband (UWB). UWB technology is frequently cited for indoor communication and localization due to its high precision. UWB implementation often incurs high infrastructure costs and requires complex calibration (Saeidi et al., 2019), making it disadvantageous for high-scale indoor service applications due to its high infrastructure overhead and hardware limitations. Unlike the proposed vision-based system, which utilizes existing ambient surfaces for ArUco marker placement, UWB requires the deployment of multiple active anchors.

Literature based on collaborative navigation is shown to be a promising technique for infrastructure-free indoor positioning, particularly for groups of pedestrians such as rescue personnel. It relies on inter-agent ranging and shared sensor data to estimate positions without pre-installed hardware (Ruotsalainen et al., 2021). However, for autonomous service robotics in structured environments like hotel corridors, this method presents several critical disadvantages like computational complexity, where managing a real-time collaborative network requires significant bandwidth and processing power for synchronization, error accumulation due to drift, requirement for multiple agents.

Navigation Beacons like Radio beacons, magnetic beacons and low energy bluetooth beacons are also under study; Radio beacons are Fixed transmitters that emit radio frequency signals for trilateration based on signal strength or time-of-arrival. They provide wide-area coverage and can penetrate non-metallic partitions (Vaščák and Savko, 2018), but it is highly susceptible to multi-path interference and signal fading in complex corridor geometries. Magnetic beacons are Sensors or emitters that utilize the Earth's localized magnetic field or artificial magnetic signatures for position mapping, it operates without line-of-sight and remain unaffected by optical conditions like smoke or low light (Sheinker et al., 2016), however it is Extremely sensitive to electromagnetic interference from structural steel, elevators, and electronic equipment. Bluetooth Low Energy (BLE) Beacons are Small, battery-powered hardware transmitters that broadcast identifier signals to nearby electronic devices using the 2.4 GHz ISM band it is low-cost, power-efficient, and easily integrated with standard mobile computer platforms (Singh et al., 2018). They are reliable only for coarse proximity; signal instability caused by human movement or metal surfaces often results in unreliable trilateration.

Another type of navigation beacon that is popular for indoor based navigations is fiducial markers, which are predefined with 2D patterns designed to be recognized by vision systems. ArUco, AprilTags and QR codes are a few types of fiducial markers that serve as ground truth reference values to reset the odometry drift. Despite their popularity, the suitability of specific fiducial types varies based on the operational requirements of the autonomous system, QR Codes are 2D barcodes designed for high data storage density can store complex URL or text data within the pattern (Bach, Khoi and Yi, 2023), however it is incompatible for high-speed robotics as they require high-resolution imagery and near-perpendicular alignment for successful decoding, making them highly sensitive to motion blur and perspective distortion during robot maneuvers. AprilTags are a visual fiducial system optimized for high localization precision and robust detection in varying light, it is highly accurate and robust against partial occlusions (Pandey et al., 2024). The detection algorithms are computationally intensive, which can lead to frame-rate drops on edge-computing devices. ArUco Markers are synthetic square binary markers based on a simplified internal matrix. It is specifically designed for real-time performance and low computational overhead, allowing for rapid detection even on resource-constrained hardware (Patrioli, 2020). They carry a smaller data payload compared to QR codes, though this is negligible for simple ID-based localization.

In this research, ArUco markers are selected over QR and AprilTags due to their superior balance of detection speed and localization reliability. Figure 1.1 shows a sample environment of Aruco markers detected by robot. Unlike radio or magnetic beacons, these markers are passive, low-cost, and resolve the spatial ambiguity inherent in repetitive hotel corridors. The selection of ArUco markers over traditional QR codes or AprilTags is a decision rooted in the fundamental physics of robot vision and real-time processing constraints. ArUco markers utilize a simplified binary matrix essentially a grid of "bits" that allows for high-speed detection even with lower-resolution camera sensors. A mathematical constraint is vital for ensuring that markers are placed within the robot's "perceptual window." Unlike QR codes, which are highly sensitive to motion blur and require high-resolution imagery for their dense data patterns, the simplified internal matrix of the ArUco marker ensures that detection remains robust even as the robot manoeuvres at operational speeds.

A significant hurdle in indoor localization is the "Kidnapped Robot Problem" and the failures of traditional Simultaneous Localization and Mapping (SLAM) in symmetrical environments. In hotel corridors, walls often lack unique visual features or "geometric landmarks," leading to scan-matching errors in LiDAR-based systems. For a robot utilizing 2D laser range finders, a long, featureless corridor appears as an infinite tunnel, causing the SLAM algorithm to lose track of the robot's longitudinal position. This visual and geometric repetition creates an "ambiguity" that can only be resolved through absolute spatial anchors. While high-end LiDARs offer precision, their cost—often exceeding the price of the robot chassis itself—makes them a non-viable option for low-cost service applications. This economic barrier necessitates a shift toward vision-based IPS that can leverage the ambient surfaces of the environment, such as walls and ceilings, to host passive markers.

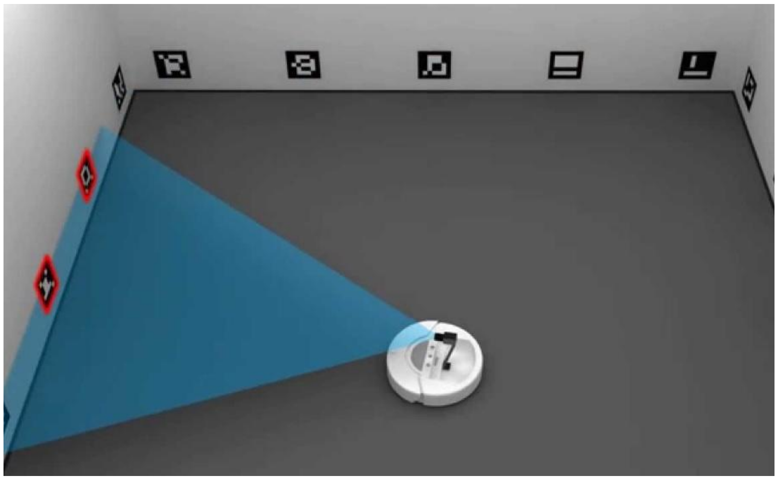


Figure 1.1- Robot detecting ArUco markers
Img src:https://roboticsbiz.com/wp-content/uploads/2023/01/robot-localization

CHAPTER 2

LITERATURE SURVEY

2.1 Path Planning and Trajectory Optimization

Path planning serves as the foundation for autonomous movement, transitioning from global exploration to local reactivity. Early foundational work by LaValle (1998) introduced Rapidly-exploring Random Trees (RRT), a method designed for high-dimensional configuration spaces. While RRT is probabilistically complete and efficient in exploration, it does not inherently guarantee the shortest path, often resulting in non-optimal routes. In contrast, Smith and Jones (2009) provided a comparative analysis of the A* and D* Lite algorithms. Their findings indicate that while A* remains superior for finding the shortest path in static environments, D* Lite is more efficient for dynamic maps due to its ability to incrementally re-plan by reusing previous computational data.

For local obstacle avoidance, the Dynamic Window Approach (DWA) was developed to provide rapid, reactive maneuvers (Fox, Burgard and Thrun, 1997). However, DWA often results in "jerky" motion because it fails to account for long-term consequences or the robot's physical smoothness. This limitation is addressed by the Time-Elastic Band (TEB) algorithm (Rost and Miller, 2014), which optimizes the full trajectory over a time horizon. TEB ensures kinodynamic feasibility by directly modeling constraints like acceleration and curvature, leading to inherently smoother and safer motion compared to traditional local planners.

2.2 Localization and State Estimation

Reliable navigation requires precise localization, typically achieved through a combination of relative and absolute sensing. Cho and Kim (2010) highlight that Dead Reckoning (DR) via wheel odometry is essential for high-frequency updates but suffers from cumulative errors that grow without bound. To mitigate this, absolute sensors are required. However, many absolute methods have significant drawbacks. Ultrasonic-based positioning is compromised by environmental fluctuations like temperature (Kim, Kim and Hwang, 2010), while radio-based systems (RFID/Bluetooth) and Wi-Fi fingerprinting often lack the sub-meter precision required for fine-grained robotic tasks (Choi and Kim, 2012; Davies and Baker, 2020).

Recent research has shifted toward visual fiducial markers. While QR codes are frequently used, they are geometrically deficient for accurate 3D pose calculation and orientation estimation (Lee, Cho and Kim, 2013). Rostkowska and Topolski (2019) demonstrated that ArUco markers offer superior robustness and higher detection success under real-world conditions like occlusion or blur. The integration of these sensors is typically managed by an Extended Kalman Filter (EKF), which Thompson and Miller (2020) validated as the optimal method for fusing intermittent absolute measurements with drifting DR data to reset cumulative errors. Furthermore, the use of wide-angle cameras to detect multiple markers simultaneously has been shown to dramatically lower measurement covariance, increasing overall system accuracy (Scott and Bell, 2021).

2.3 Integrated Systems and Industrial Applications

Modern research emphasizes the integration of these components into robust architectures for unstructured or industrial environments. Scene perception-based visual navigation allows robots to understand contextual details for better obstacle avoidance, though it requires significant computational power (Ran, Yuan and Zhang, 2021). In cost-sensitive scenarios, WiFi-beacon datasets have been developed to facilitate 3D localization, although they remain susceptible to signal fluctuations (Abu Kharmeh, 2023).

In manufacturing settings, the integration of visual-based odometry and perception modules on omnidirectional Automated Guided Vehicles (AGVs) provides robustness against wheel slip and textureless floors (Patruno et al., 2024). Neaz, Lee and Nam (2023) further demonstrated that combining multiple LiDAR sensors with G-mapping SLAM and a layered costmap planner within a ROS-based architecture allows for adaptive response to dynamic obstacles in real logistics settings. Finally, the importance of long-term autonomy is underscored by Portugal, Araujo and Couceiro (2021), who proposed an incremental development strategy. Their findings suggest that a layered guidance system with redundancy and fault-detection recovery logic is essential for persistent service robots, enabling hundreds of hours of operation with minimal human intervention.

2.4 Strategic Marker Placement and Configuration

Beyond basic algorithm selection, the physical deployment of artificial markers—such as ArUco, AprilTag, and QR codes—is a critical determinant of navigation success. A systematic mapping study by Alghamdi, Al-Marakeby and Abdel-Mageid (2025) emphasizes that localization robustness is not merely a function of marker count, but depends heavily on visibility conditions, lighting, and environmental-specific placement strategies. Specifically, Hinderer, Scheffler and Yang (2025) investigated planar layouts, finding that marker orientation and inter-marker spacing directly influence pose estimation accuracy. Their research suggests that optimizing layouts for specific structural features like corridors is more effective than simply increasing the number of markers to prevent localization instability.

A consistent finding across the literature is the diminishing return associated with dense marker deployment. Mantha and Garcia De Soto (2022) demonstrated through simulations that while marker size and height are significant, excessive markers provide minimal improvement to navigation success. Instead, markers should be placed at regular, strategically calculated intervals. Supporting this, Beinhofer et al. (2013) formulated landmark selection as an optimization problem, proving that a "greedy" selection of specific landmark locations significantly reduces pose uncertainty while requiring fewer landmarks compared to random or uniform distribution.

Furthermore, the geometry of marker placement plays a vital role in high-precision tasks. Basso (2020) compared planar and multi-plane (3D) configurations, concluding that 3D arrangements notably improve pose estimation accuracy, which is particularly useful for tasks like autonomous docking. To streamline this planning, Kayhani, Schoellig and McCabe (2023) proposed a perception-aware optimization framework using the Fisher Information Matrix (FIM). This mathematical approach ensures high localization accuracy with a minimal, cost-effective infrastructure by placing fewer but more informative tags.

2.5 Research Gap and Novelty

A prominent research gap is the lack of deterministic safety at junctions for vision-based systems, which often lose localization during high-speed turns. This work closes this gap by mandating dual-tag placement at every intersection. Another significant gap is the high infrastructure of redundancy and visual clutter associated with dense marker deployment in long, symmetrical corridors.

The novelty of the work lies in the development of a structured, visibility-aware optimization framework that automates the transition of manual observations to effective robotic navigation deployment in indoor geometries. Existing literature has pointed out that uniform marker distribution is not efficient, and the study shows the same. The work also categorizes beacons based on environmental priority. This approach uses mathematical feasibility analysis and employs a greedy selection technique with the observed inference. This pattern maximises the robot's lateral field of view and, more effectively than standard structured layouts, minimises the Maximum Blackout Length. Finally, the study establishes a three-step validation that combines theoretical MATLAB modelling, high-fidelity Webots digital twin simulations, and real-world implementation on an NVIDIA Jetson Nano-powered Yahboom Jetson bot, ensuring that the system's efficiency is maintained and compared across both virtual and physical domains.

2.5 Problem Definition

Optimize the placement and number of ArUco tags to ensure reliable indoor robot navigation with minimal infrastructure

2.6 Objectives

To achieve the problem statement, the following objectives are drafted-

a) To calibrate the onboard camera system to determine the size and ensure accurate ArUco markers
b) To configure the functionally specific markers and feasibility analysis for marker placement
c) To design a hierarchical optimization technique for the placement of ArUco markers by analyzing spatial constraints, visibility requirements, and navigation performance.
d) To validate the proposed technique using simulation and in a real-time environment.

CHAPTER 3

METHODOLOGY

3.1 Proposed Methodology-

A multi-phase approach to optimize the deployment of navigation beacons (ArUco markers) for autonomous mobile robots is implemented, Figure 3.1 shows the overall process flow of the work. An environment is set up in Step 1 according to the problem defined, which consists of repetitive structures. Tags are placed manually in steps 2 and 3, following the constraints, and inferences are noted, which is explained further in subsection 3.3. Step 4 chooses the optimal tag size considering the camera parameters and environment parameters. Step 5 consists of the design of an algorithm based on the inferences collected, and the algorithm is validated for its reliability in Steps 6 and 7.

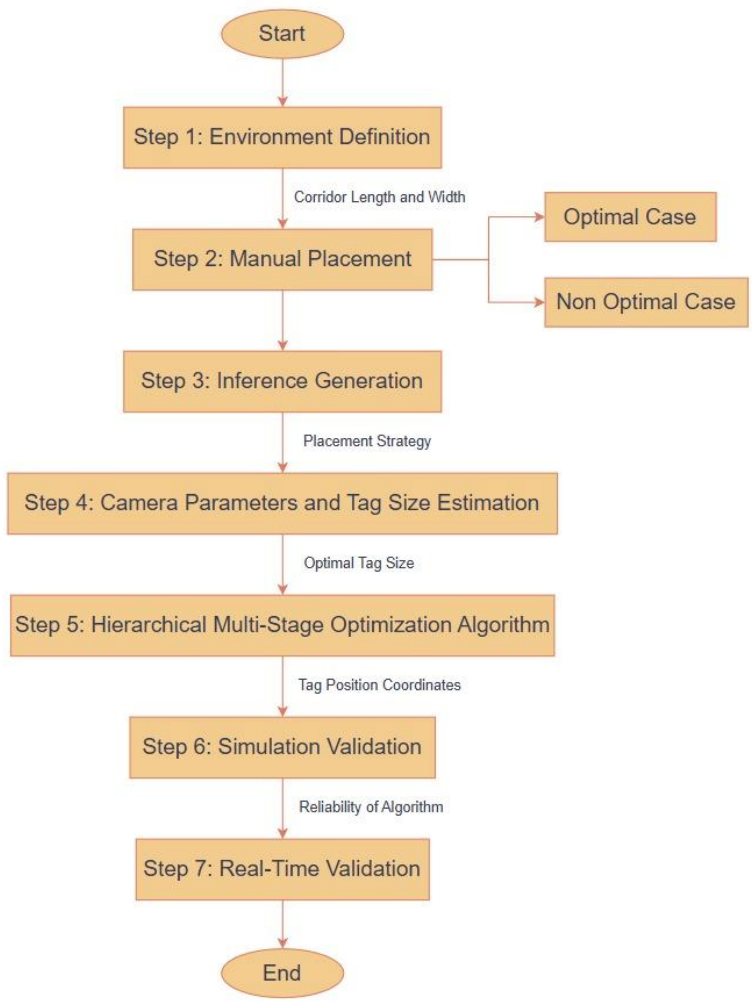


Figure 3.1- Flowchart of Proposed Methodology

3.2. Manual Placement and Inference generation -

The beacons are categorized into three types based on placement locations, i.e. Functional Tags, which are placed at doors or task areas, Junction Tags, which are placed at intersections, and Corridor Tags for straight-line navigation.

Visibility constraints are derived using the Environment and Camera parameters and are used as baseline constraints for the manual placement technique. The cases are analyzed under Optimal and Non-Optimal configurations. Figure 3.2 shows the methodology for manual placement.
Non-optimal Cases – Tags are placed uniformly with an interval of 25cm single side and alternating sides of the corridor walls.
Optimal Cases – Deployment follows specific visibility constraints, testing configurations – one side, alternate side, and a randomized optimal placement; Figure 3.3 shows the manual placement of the markers.

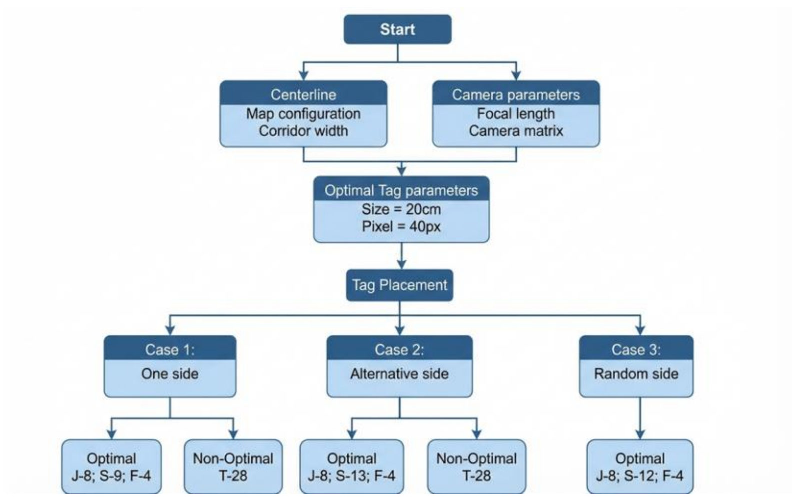


Figure 3.2- Methodology for manual tag placement to maximize localization availability.

The inferences noted from manual placements are that optimality is highly sensitive to tag dimensions(s); smaller tags provide insufficient localization, while large tags create over-dense results that clutter the camera's field of view (FOV). Therefore, an optimal tag size that suits the camera requirements and the environment is required.

Uniform placement yields high redundancy (>40%), which is useful for obscured tags but inefficient for infrastructure requirements. Junction and functional tags are mandatory for deterministic navigation. Pre- and Post–junction visibility is critical for re-localization and reduction of localization failures. Alternating placement is more effective for wide corridors, whereas single-side placement is sufficient for narrow paths.

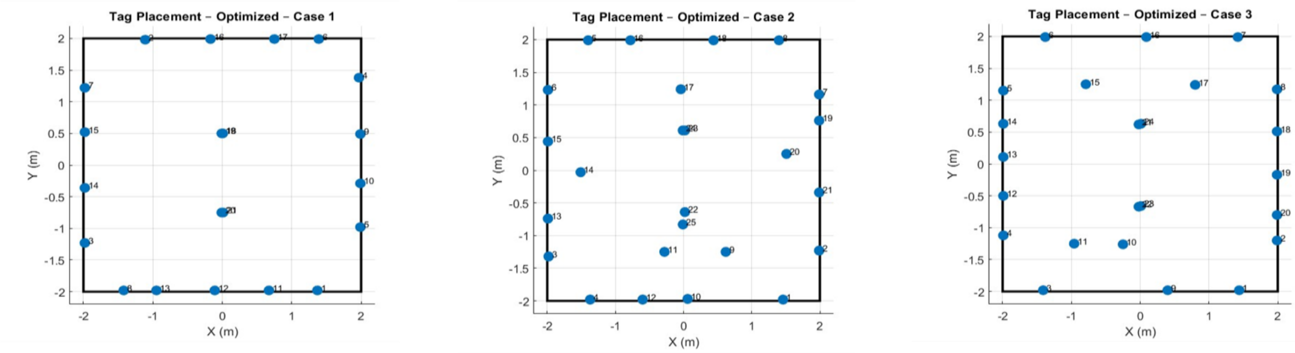


Figure 3.3 – MATLAB-generated plot of three optimal cases of manual marker placement.

3.3 Proposed Optimization Technique -

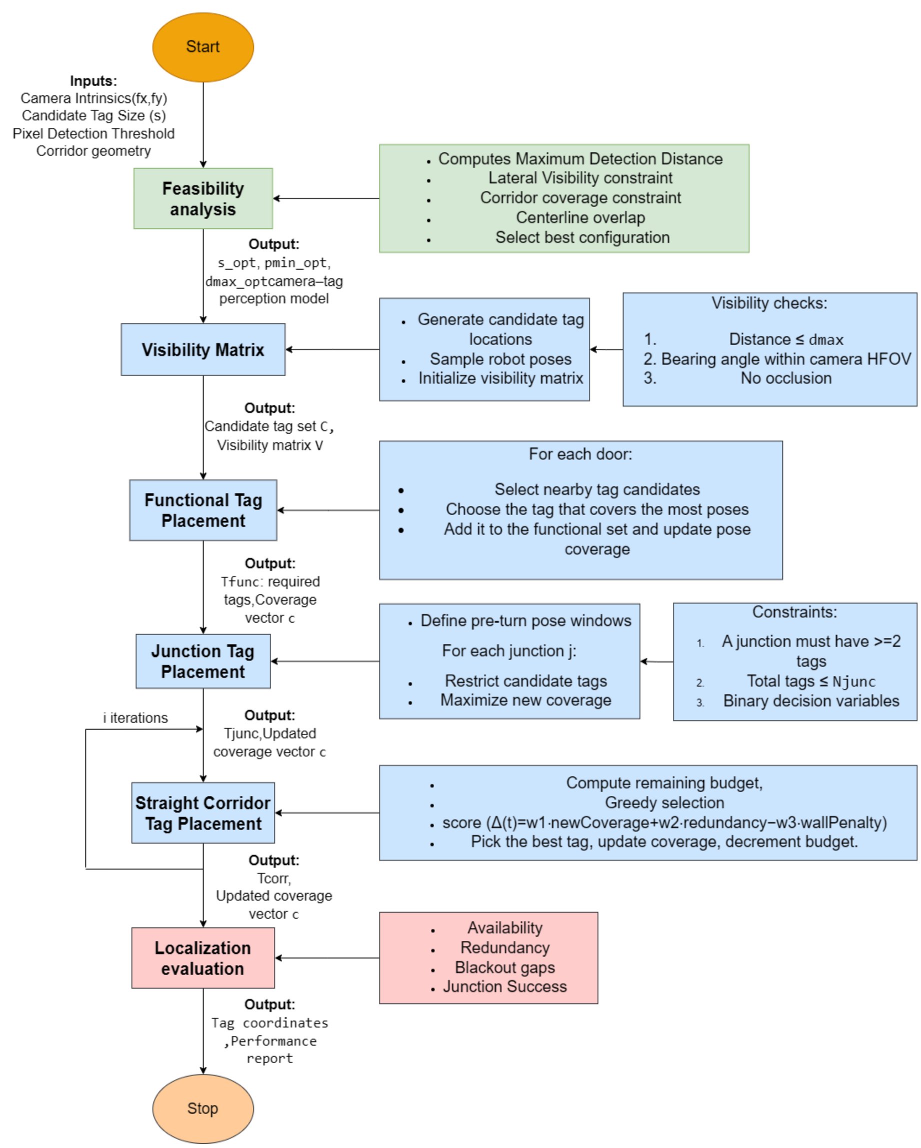


Figure 3.4- Proposed Hierarchical optimisation algorithm

A hierarchical optimization technique is proposed, as shown in Figure 3.4 to maximize tag visibility, which ensures robot localization reliability within constrained indoor environments. Initially, feasibility analysis using camera intrinsics and corridor geometry is carried out, which results in a robot perception model. Visibility matrix to map candidate tag locations against all the sampled robot poses is generated. An optimal tiered strategy is employed that prioritizes high-value areas, specifically targeting functional zones like doorways and critical junctions where manoeuvring requires high tag density. Other locations are addressed using a greedy selection algorithm that balances coverage and redundancy against the infrastructure costs.

3.4. Optimization Framework

3.4.1 Camera parameters and Tag size estimation

The objective is to select a minimal subset of ArUco tags S from a candidate pool C, such that the robot's path P achieves maximum localization availability while maintaining redundancy. The constraints are

1. Detection feasibility - A tag is only selected if its distance d satisfies as shown in Eq. 3.1

$$\frac{f_s}{P_{x,max}} \leq d \leq \frac{f_s}{P_{x,min}} \tag{3.1}$$

2. Geometric visibility – The bearing angle relative to the robot must satisfy, and Eq. 3.2 shows the formula

$$\text{Angle} \leq \frac{\text{HFOV}}{2} \tag{3.2}$$

3. Mandatory placement – Functional and junction tags to be selected, as shown in condition eq.3

$$x_j = 1 \quad \text{if } A_j \in F \cup J \tag{3.3}$$

A multi-stage optimization algorithm, as shown in Figure 3.4, is designed to Optimize the placement and number of ArUco tags to ensure reliable indoor robot navigation with minimal infrastructure. Minimal infrastructure refers to the trade-off between the reduction of ArUco tags while increasing the localization availability.

The algorithm utilizes camera intrinsics (fx, fy) and pixel thresholds (PxThr) to calculate the maximum detection distance (dmax), as shown in Eq. 3.4. A visibility matrix(V) is derived by sampling potential robot poses along the centerline of the robot path and checking for distance, bearing angle and occlusion constraints.

A configuration is feasible only if it satisfies two environmental constraints

1. Lateral visibility: d_{max} must exceed the maximum wall offset to ensure the robot can visualize the side walls from the corridor centerline
2. Corridor coverage: d_{max} must exceed the maximum corridor width to ensure continuous coverage.

$$D_{max} = \frac{f \cdot s}{p_{min}} \tag{3.4}$$

Candidate tags (C_j) are generated along all wall surfaces with a uniform spacing of 0.15m, from which the set of tags that pass the visibility constraints, as shown in Eq. 3.5, for every robot pose (Pi) are stored, a binary visibility state is calculated:

$$V_{i,j} = 1\ if(dist \leq d\_\{max\}) \cap (\,|bearing| \leq HFOV\,/\,2)\ \cap\ (n \cdot v > 0) \tag{3.5}$$

3.4.2 Multi-Stage Tag Placement

The algorithm is divided into three sequential stages,

Stage 1 Functional Stage – The algorithm initially prioritizes high-priority task areas. Candidate tags that are closest to the doorways are selected to ensure task-specific localisation.

Stage 2 Junction stage – 2-4 tags are selected for junction-based localization. Robot localization during Pre-turn and post-turn is mandatory since it leads to localization errors, and instant relocalisation is mandatory. L- and T- based intersections are considered as junctions.

Outer tags – Placed at a calculated distance to maximize the viewing angle during the turn.

Inner tags – placed closer to the junction centre to provide a secondary signal during the maneuver.

Stage 3 Corridor stage – The remaining straight corridors are filled using a greedy iterative algorithm that selects tags from the candidate pool based on a weighted objective function, as shown in Eq. 3.6

$$\text{Score} = \text{NewCoverage} - \text{Penalty} \tag{3.6}$$

A penalty is applied if a candidate is placed on the same side of the wall as previously selected It selects candidate tags that minimize the blackout lengths while avoiding unnecessary redundancy, since the straight corridors are less error prone.

Overall, each stage of the algorithm contains its own optimization goal, i.e functional stage requires optimization of task points and base localization is improved with the algorithm. Junction stage improves navigation safety with the utilization of mandatory pre and post turn success. The straight corridor stage reduces the gap while increasing the total localization availability for the robot. The pseudo-code of the algorithm is given below.

Algorithm 1: ArUco Placement Optimization

Input: Camera Intrinsics K, Tag Sizes s, Pixel Thresholds P_{min}, Environment Map M

Output: Optimized Tag Set T_{final}, CSV

1. Phase I: Feasibility Analysis
    a. For each s and P_{min} in the candidate list:
        i. Calculate $d_{calc} = f \,.\, s \,/\, P_{min}$.
        ii. Check Constraints:
            1. d_{calc} >=1.0 m (Lateral Visibility).
            2. 2. d_{calc} >=2.0 m (Corridor Width) .
        iii. Score configurations and select Optimal Config s = 0.20 m, P_{min} = 40 px, d_{max} = 1.1 m) .
2. Phase II: Visibility Matrix Construction

a. Initialize Matrix V of size (RobotPoses x TagCandidates).
b. For each pose P_i and candidate C_j:
    i. Set $V_{i,j}$ <- 1 if C_j is within d_{max}, inside HFOV, and oriented toward P_i .

3. Phase III: Multi-Stage Hierarchical Selection
    a. Stage 1 (Functional): Select candidates closest to the 4 door targets.
    b. Stage 2 (Junction): For each corner, place 2 tags on the horizontal leg and 2 on the vertical leg using fixed offsets.
    c. Stage 3 (Straight): While path is not fully covered:
        a. For each remaining candidate j:
            i. Calculate Gainj = sum(uncovered poses visible by j)
            ii. Calculate penalty j = 100 if side j matches the nearest neighbour in S.
            iii. Calculate Score = NewCoverage - Penalty.
            iv. Iteratively add the highest-scoring candidate to T_{final}.
4. Phase IV: Export
    a. Apply +90 deg yaw correction for Webots axis alignment.
    b. Generate a CSV file with ID, Type, and X, Y, Z, Yaw

3.5 Environmental Setup

A hotel-like indoor corridor environment characterized by repetitive structures is designed and set up, as shown in Figure 3.5. The environment is set up with functional areas such as doors and junctional zones, which are error-prone zones and have a higher possibility for localization failures. Constraints are defined by measuring the corridor width to determine the lateral visibility and the required bearing angles for pre-turn and post–turn junction availability.

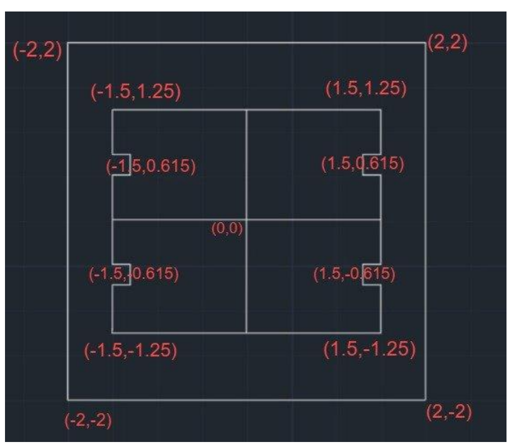


Figure 3.5- AutoCAD drawing of the environment marked with coordinates

The optimized coordinates are imported into the 3D simulation environment. Figure 3.6 shows the Webots interface with the setup environment. The robot's performance is monitored in real-time to verify that the tags placed satisfy the navigation requirements in dynamic settings.

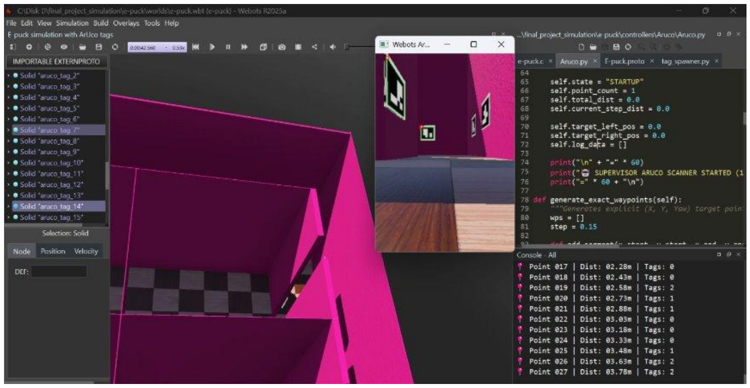


Figure 3.6- Simulation environment in Webots

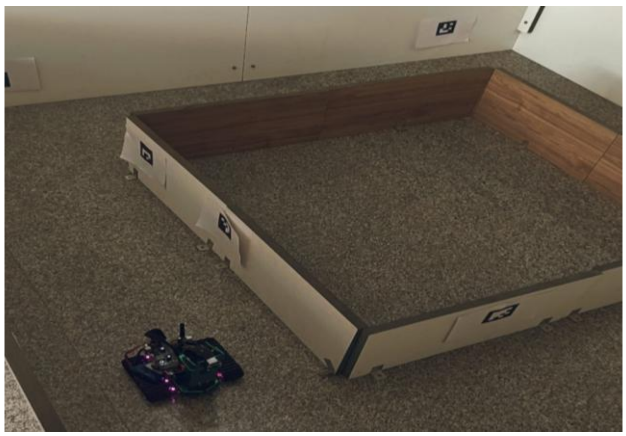


Figure 3.7- Yahboom Jetson bot and Environemental setup

The validation uses the Yahboom Jetson bot(shown in Figure 3.7), with a scaled-down environment as shown in Figures 3.5 and 3.6, an AI-driven smart robot based on the NVIDIA Jetson Nano 4GB (B01). The bot enables real-time execution of the ArUco detection and localization stack. Camera: Sony IMX335 5MP USB Camera, featuring an 8MP IMX219 chip with a maximum resolution of 3280 × 2464. The features and their specifications are listed in Table 3.1.

Table 3.1 Yahboom Jetson bot specification

Parameter	**Specification**
Max Resolution	2592 x 1944 P
Field of View (FOV)	175°
Focal Length	3.65 mm
Size	25mm x 24mm

CHAPTER 4

RESULTS AND DISCUSSION

The algorithm was evaluated across various scenarios and maps, using different scaling factors to verify its reliability. The results were consistently similar across all maps and case studies. One representative case scenario, along with its corresponding map, is presented in this paper. Figure 4.1 shows the process and metrics used for evaluation of the model across the three-step validation

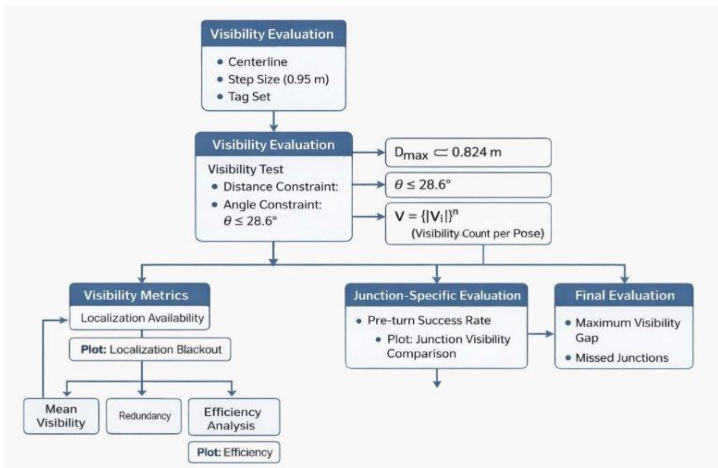


Figure 4.1- Flowchart for Evaluation

4.1 Manual Placement: Optimal vs. Non-Optimal

Tags were placed at a uniform density of 25cm on both single and alternating sides of the wall. This yielded a high redundancy of 45% but resulted in excessive infrastructure with only 77% availability, as shown in Table 4.2. The three optimal cases: single side(C1), alternate side(C2), random side(C3).

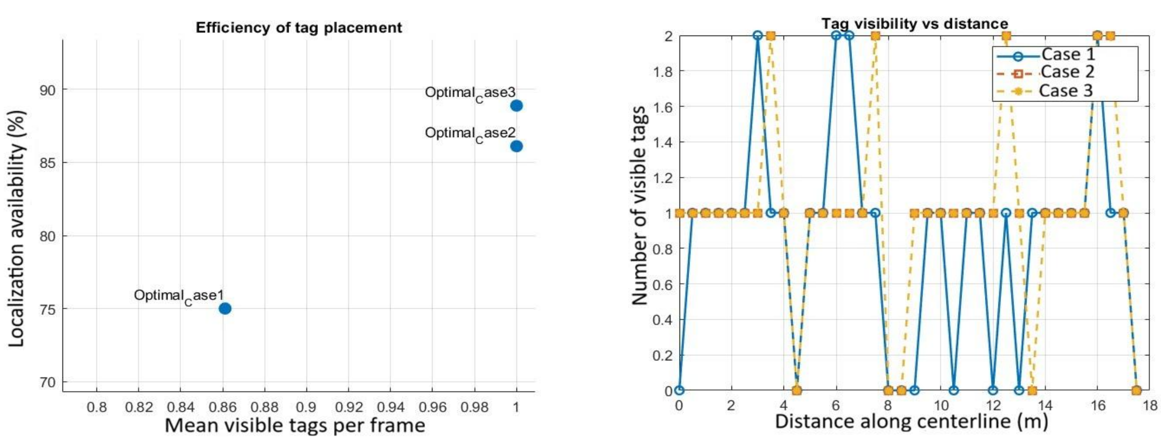


Figure 4.2(a) efficiency of tag placement Results, 4.2(b) Tag visibility along the path travelled

Optimal case 3 acheived the highest performance with 89%(Figure 4.2(a)) availability from Table 4.1 and 75% junction success rate as shown in Figure 4.2(b). It maintained low redndancy(11%) and the smallest maximum visibility gap of 0.5m.

Table 4.1 Overall results of Manual placement

Case	**Availability(%)**	**Junction Avail(%)**	**Redundancy(%)**	**Missed Junction**
Optimal C1	75	56.25	11.11	7
Optimal C2	86.11	68.75	13.88	5
Optimal C3	88.88	75	11.11	4
NonOpt C1	77.77	68.75	44.44	2
NonOpt C2	77.77	68.75	30.55	2

4.2 Results obtained from the optimization algorithm

Table 4.2 Feasibility Comparison across all the possibilities

Tag (m)	**Pixcel Threshold**	**MaxDist (m)**	**Status**
0.1	30	0.73	Fail
0.1	40	0.55	Fail
0.1	50	0.44	Fail
0.15	30	1.1	OK
0.15	40	0.82	Fail
0.15	50	0.66	Fail
0.2	30	1.46	OK
0.2	40	1.1	OK
0.2	50	0.88	Fail

The Line graph in Figure 4.3 shows that as the pixel threshold increases, the maximum detection distance decreases for ArUco tags of different sizes (0.10m, 0.15 m, and 0.20 m). The algorithm feasibility analysis selected an optimal configuration of 0.2m tags with a 40px threshold, providing a maximum detection distance (dmax) of 1.1m. (Table 4.2).

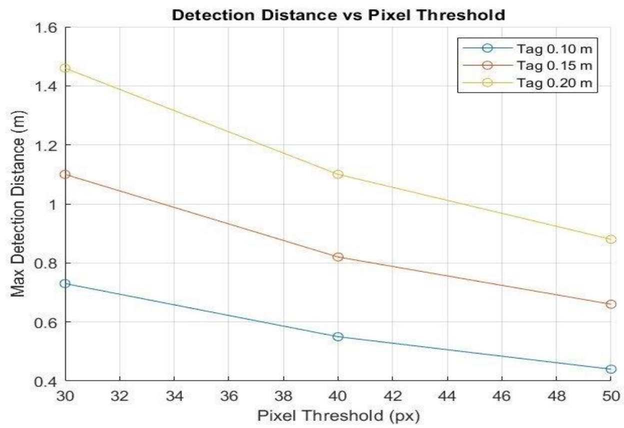
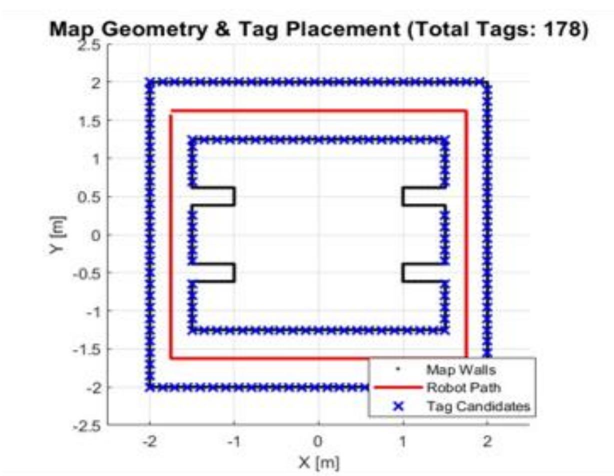
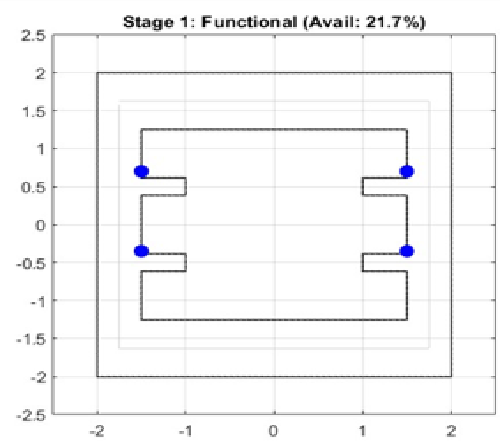


Figure 4.3- Detection distance vs Pixel threshold for different tag sizes

A set of candidate tag locations is setup; candidate tags are placed 15cm from one another, and a total of 178 tags is generated as shown in Figure 4.4(a). The algorithm prioritizes door notches and task specific zones. In the environment provided, 4 doors are present, and so 4 functional tags are selected as shown in Figure 4.4(b), and a 97.8% reduction from candidates' tags while achieving an overall localization availability of 21.7%





Figures 4.4(a) Total generated candidate tags, 4.4(b) Chosen functional tags from the set

Junction tags were added to ensure deterministic navigation are critical turns. There were 4 junctions present at each corner of the environment, thereby adding an additional 16 tags, visualized in Figure 4.5 increased the availability to 60% and reduced the blackout length to 2.3m

The remaining gaps were filled using a greedy heuristic algorithm. Over 12 iterations, the system evaluated the candidate pool and reached a stable coverage point. The final configuration used 32 tags, which gave 82% reduction to achieve 94.2% localization availability(from Table 4.3. Tag placements can be visualized in Figure 4.6, with a maximum blackout length of 0.3m.

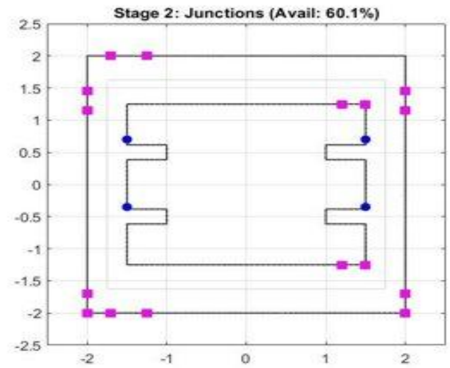


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Figure 4.5- Chosen Junction tags from the total generated candidate tag set

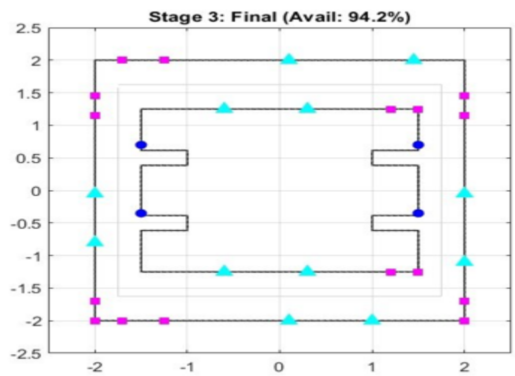

Figure 4.6 Final allocation of corridor tags

Table 4.3 Overall results obtained across all stages

Stage	**Tags Used (%)**	**No.Tags Used**	**Total Candidates**	**Reduction (%)**	**Localization Availability (%)**	**Total Blackout Length(m)**
Functional	2.2	4	178	97.8	21.7	4.9
Junction	11.2	20	178	88.8	60.1	2.3
Straight	18	32	178	82	94.2	0.3

Figure 4.7(a) shows the localization availability through mean tags visible per frame across the path travelled by the robot, and Figure 4.7(b) points out the drastic reduction of blackout across the stages of the algorithm

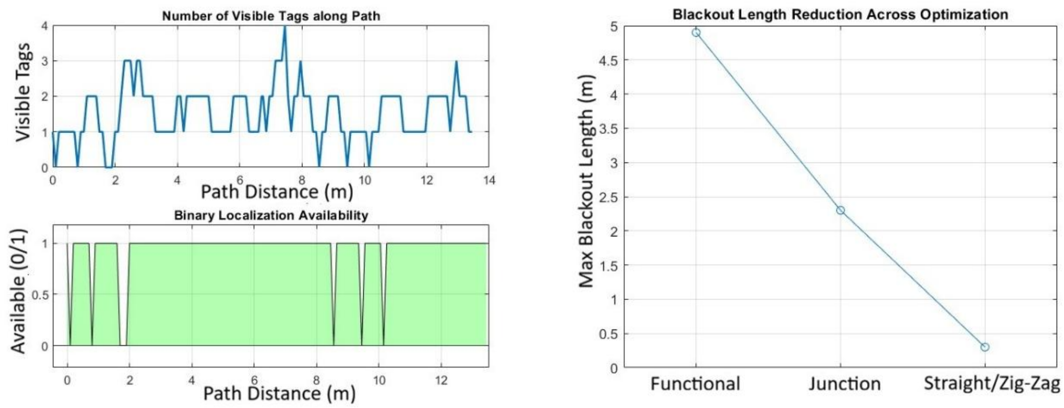

Figure 4.7(a) - Localization availability plots, 4.7(b) Blackout across stages

4.3 Simulation Validation

The optimized coordinates were imported into Webots R2025a environment. The robot traversed a total distance of 14m with a mean visibility of 1.011 tags per frame, Figure 4.8(b). A localization reliability of 85% was received(Figure 4.8(a)), with a 100% pre-turn success at junctions from Table 4.4. Blackouts were localized to short, manageable intervals(Figure 4.8(c)); Therefore the results confirm the ability to maintain stable navigation.

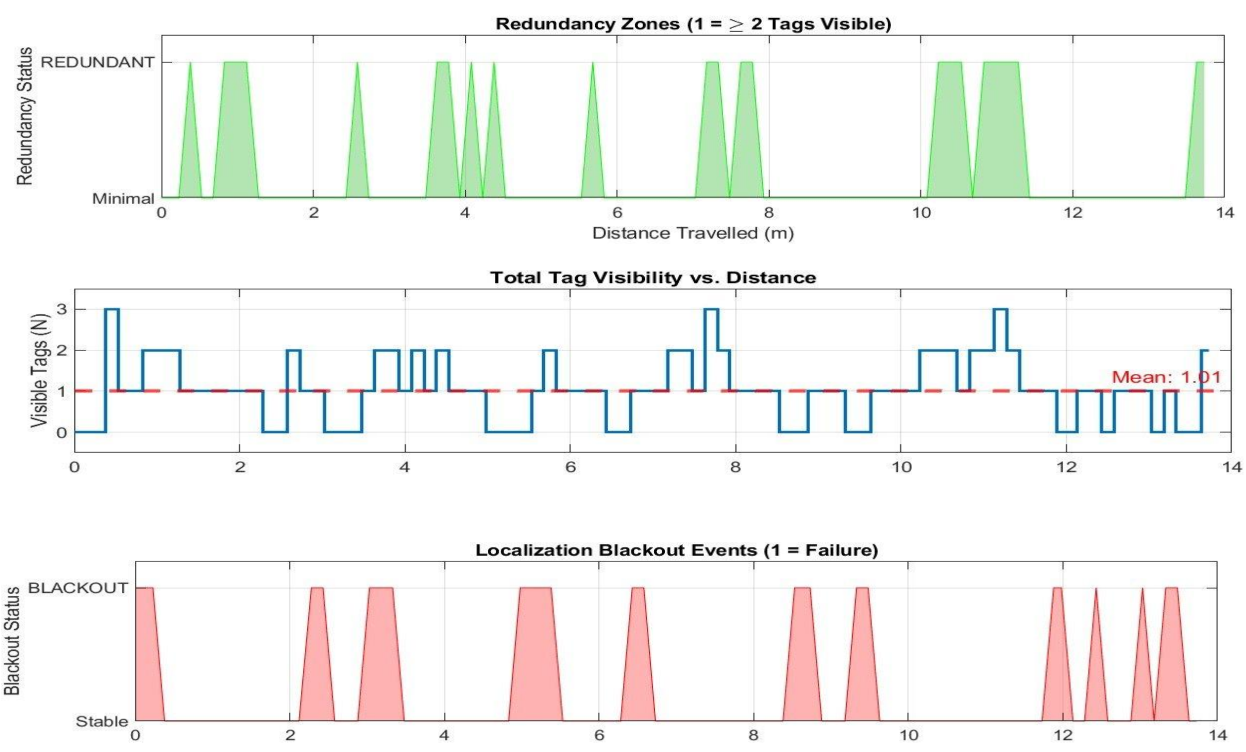

Figure 4.8(a) Redundancy plot, 4.8(b) Tag visiblity graph, 4.8(c) Blackout regions

Table 4.4 Overall results from Webots Simulation

Metric	Value
Total Distance Travelled	13.73 m
Mean Tags Visible	1.011
Max Tags Observed	3
Localization Reliability (Path with ≥ 1 Tag)	84.58%
Blackout Rate (Failure) (Path with 0 Tags)	16.32%
Redundancy Rate (Optimal) (Path with ≥ 2 Tags)	24.21%

4.4 Real-Time validation(Hardware)

The proposed algorithm was evaluated in a physical environment using a scaled-down representation of the map shown in Figure 3.5. The experimental setup accounted for the specific dimensions and kinematic constraints of the Yahboom Jetson bot robot. A feasibility analysis determined that a 10 cm ArUco tag size was optimal, given the environment's scale and the camera's intrinsic parameters. Figure 4.9 illustrates the optimized tag placement strategy generated by the MATLAB-based algorithm. While the theoretical model predicted a localization availability of 94%, physical validation yielded a localization availability of 92% (from Table 4.5 and Figure 4.10). This close alignment between simulated and empirical data confirms the validity of the algorithm's underlying hypotheses.

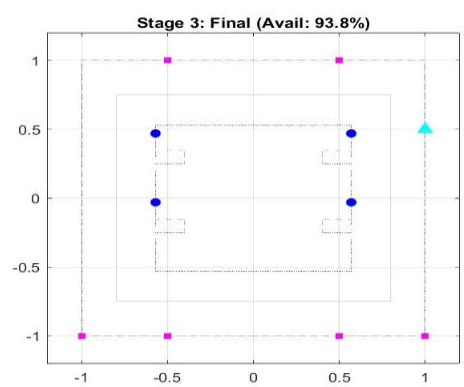

Figure 4.9- Optimized tag placement for Hardware validation

Table 4.5 Evaluation results from Hardware

Metric	Value
Total Frames	2250
Blackout Percentage	1.32%
Availability Percentage	98.6%

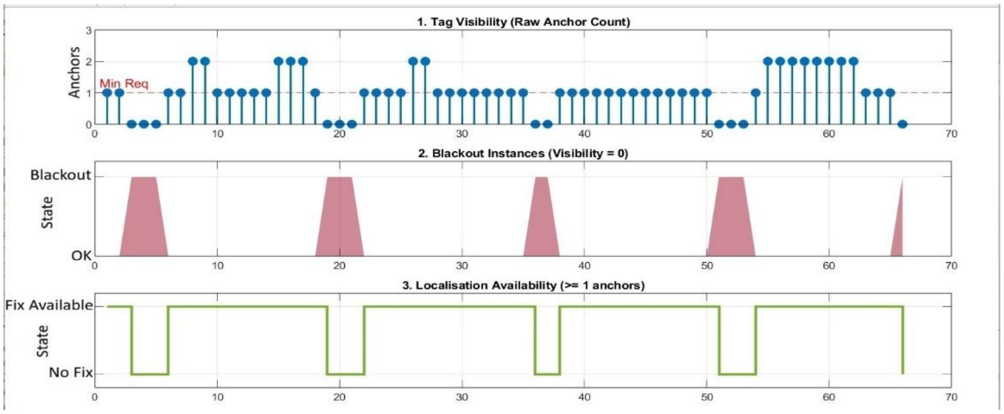

Figure 4.10- Evaluation plots of the result table

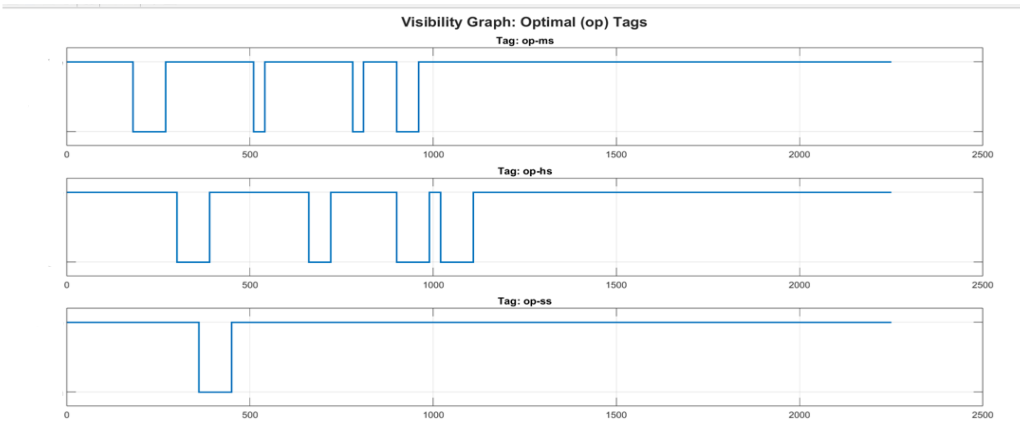

Figure 4.11(a)- Speed-based evaluation over Optimal placement of tags

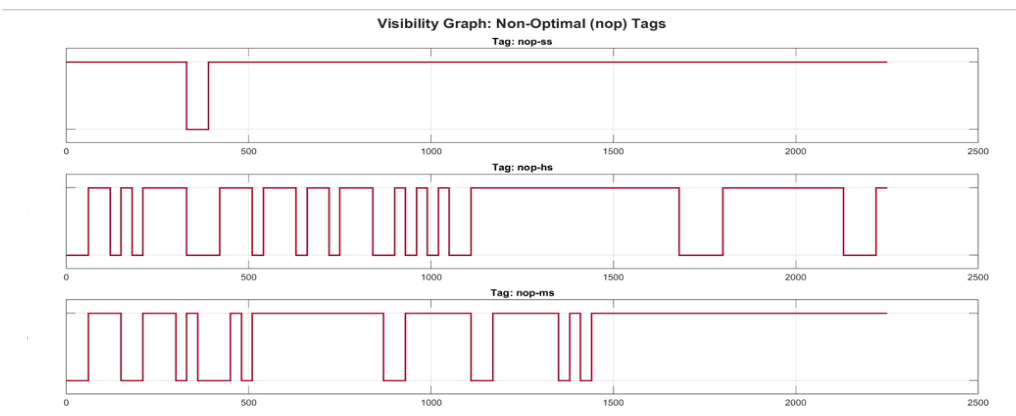

Figure 4.11(b)- Speed-based evaluation over Non- Optimal placement of tags

Furthermore, the detection accuracy was tested by varying the speed of the robot and placements of tags, it can be seen from Figures 4.11(a) and 4.11(b) (ss- slow speed 5.54cm/s, ms - 8.71cm/s, hs- high speed 15.24cm/s)the detection accuracy decreases as the speed increases and more blackout regions are found. A higher resolution camera can be utilized with tasks that require fast moving vehicles.

The results demonstrate that uniform tag placement is inefficient and is infrastructure heavy. Manual testing confirmed that alternating placement provides effective coverage–to-tag ratios compared to single-side deployement. The algorithm successfully automated the inference by prioritizing error-prone areas like the junctions and task points before filling out the corridor gaps, maintaining deterministic coverage. By using the Greedy selection algorithm, the system achieved 94.2% availability. This strategy directly addresses the primary challenge of indoor navigation, that is resolving visual ambiguity in a repetitive environment with minimal cost. The 100% success rate in pre-turn for junction is critical, as it ensures the robot reliably detects turning beacons before losing its line of sight during heading changes.

The results obtained through both Webots simulations and physical hardware validation consistently support the algorithm's hypotheses across various environmental scenarios. The high localization availability achieved in real-world testing demonstrates the algorithm's efficiency and robustness. Furthermore, the successful transition from simulation to hardware proves that this optimized tag placement strategy is viable for deployment in complex, real-world indoor environments, such as hotel corridors or repetitive industrial spaces.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The work presents a cost effective, vision-based navigation system that optimizes ArUco marker placement, through a hierarchical three-stage optimization algorithm, the study transitioned from inefficient unfirm beacon distribution to a visibility aware deployment. By automating the inferences received from manual placement, the proposed algorithm achieved a localization availability of 94.2%, while reducing the total infrastructure by 82%. Validation across MATLAB, Webots, and hardware demonstrated a 100% pre-turn success rate at junctions and a mean visibility of 1.011 tags per frame. Therefore, integrating camera intrinsics and environmental geometry into marker placement enables reliable autonomous navigation with minimal infrastructure. These findings offers an accessible framework for deploying service robots in real-world applications, such as hotel and industrial corridors.

5.2 Future Scope

The integration of sensor fusion techniques, specifically utilizing Extended Kalman Filters (EKF) to combine visual beacon data with IMU and wheel odometry, which is expected to eliminate the residual blackout rate observed during sharp cornering maneuvers. To improve robustness, further studies will model dynamic obstacles such as pedestrians or other mobile units to maintain localization during temporary occlusions of ArUco markers.

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